

MTH100 Paper Question

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Regards By VUWAYS Team

Welcome To All New Members

Short-Answer Questions. Questions 1 and 2 are short-answer questions. Put your answer in the box provided. Full marks will be given for a correct answer placed in the box. Show your work also, for part marks. Each part is worth 2 to 3 marks, and not all parts are of equal difficulty. **Simplify your answers as much as possible in Questions 1 and 2.**

2 marks

1. (a) Let $y = 10^{1-x^2}$. Find $\frac{dy}{dx}$.

Answer: $\frac{dy}{dx} = -2 \ln(10)x \cdot 10^{1-x^2}$

Solution: Use the chain rule

$$\frac{dy}{dx} = \ln(10) \cdot 10^{1-x^2} \cdot (-2x)$$

2 marks

- (b) Evaluate $\tan^{-1} \tan \frac{11\pi}{4}$.

Answer: $-\frac{\pi}{4}$

Solution: As done in class, one way to solve this problem is to draw a unit circle and mark down the angle of $\frac{11\pi}{4}$ in the second quadrant. Shifting the angle back into the range of $[-\frac{\pi}{2}, \frac{\pi}{2})$ yields $-\frac{\pi}{4}$ because $\tan x$ is negative in the second and fourth quadrant.

2 marks

- (c) $y = (\sin x)^{\ln x}$. Find $\frac{dy}{dx}$.

Answer: $y' = \left(\frac{\ln(\sin x)}{x} + \ln x \frac{\cos x}{\sin x} \right) y$

Solution: Use logarithmic differentiation.

$$\begin{aligned} \ln y &= \ln x \cdot \ln(\sin x) \\ \frac{1}{y} y' &= \frac{1}{x} \cdot \ln(\sin x) + \ln x \cdot \frac{1}{\sin x} \cdot \cos x \end{aligned}$$

Multiplying through by y gives the answer in the box.

3 marks 2. (a)

Simplify $\sin(\sec^{-1} x)$. Note: $\sec^{-1} x$ is the inverse function of $\sec x$.

$$\text{Answer: } \frac{\sqrt{x^2-1}}{x}$$

Solution: Let $y = \sec^{-1} x$ then $\sec y = x$ or equivalently $\cos y = \frac{1}{x}$. Drawing a triangle with angle y as done in class gives $\sin y = \frac{\sqrt{x^2-1}}{x}$.

3 marks (b) Find all x -coordinates at which the tangent line to the curve $x^2y^2 = 16$ has slope -1 .

$$\text{Answer: } x = \pm 2$$

Solution: Implicit differentiation gives

$$2xy^2 + x^2 \cdot 2yy' = 0.$$

Plugging in $y' = -1$ and factoring gives

$$2xy(y - x) = 0.$$

This implies $x = y$ since there are no points on the curve with $x = 0$ or $y = 0$. The curve's equation thus becomes $x^4 = 16$ or $x = \pm 2$.

3 marks (c) Which of the following functions satisfies $\frac{dp}{dt} = p^3$?

A) $p(t) = \frac{1}{4}t^4$

B) $p(t) = -\frac{1}{t+3}$

C) $p(t) = \frac{1}{\sqrt{1-2t}}$

Answer: C)

D) None of these.

Solution: By differentiating each expression for $p(t)$ separately, we conclude that C) is the correct answer, i.e.,

$$\frac{dp}{dt} = -\frac{1}{2(1-2t)^{3/2}} \cdot (-2) = \frac{1}{(1-2t)^{3/2}} = \left(\frac{1}{1-2t}\right)^{3/2} = p^3$$

3 marks (d) Find the absolute maximum and absolute minimum values of $y = x + \frac{1}{x}$ on $[0.2, 4]$.

Answer: max. value is $y(0.2) = 5.2$, min. value is $y(1) = 2$

Solution: Use the closed interval method. Start by finding critical numbers by checking where the derivative equals zero or isn't defined:

$$y' = 1 - \frac{1}{x^2} = 0$$

So $x = 0, \pm 1$. Ignore 0 and -1 since they aren't in $[0.2, 4]$. We compare

$$y(0.2) = 0.2 + 5$$

$$y(1) = 2$$

$$y(4) = 4 + 0.25$$

to get the answer in the box.

3 marks (e) Write down the degree 2 Taylor polynomial of $f(x) = \tan x$ at $x = \pi/4$.

Answer: $T_2(x) = 1 + 2(x - \pi/4) + 2(x - \pi/4)^2$

Solution: We need to determine

$$T_2(x) = \sum_{k=0}^2 c_k (x - \pi/4)^k$$

where $c_k = \frac{f^{(k)}(\pi/4)}{k!}$. We have

$$f^{(0)}(x) = \tan x \quad f^{(1)}(x) = \frac{1}{\cos^2 x} \quad f^{(2)}(x) = 2 \frac{\tan x}{\cos^2 x}$$

and so

$$c_0 = 1 \quad c_1 = 2 \quad c_2 = 2$$

which gives the answer in the box.

Full-Solution Problems. In questions 3–5, justify your answers and **show all your work**. If a box is provided, write your final answer there. Unless otherwise indicated, **simplification of answers is not required in these questions**.

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3 marks 3. (a) Use a suitable linear approximation to estimate the value of $\sqrt[3]{26^2}$. Express your answer

as a single fraction.

Answer: $\frac{79}{9}$

Solution: Let $f(x) = x^{2/3}$ and expand around $x = 27$. Then

$$f(27) = 9$$

$$f'(x) = \frac{2}{3}x^{-1/3}$$

$$f'(27) = \frac{2}{9}$$

$$\begin{aligned} T_1(x) &= f(27) + f'(27) \cdot (x - 27) \\ &= 9 + \frac{2}{9}(x - 27) = \frac{2}{9}x + 3 \end{aligned}$$

and thus

$$T_1(26) = 9 + \frac{2}{9} \cdot (-1) = \frac{79}{9}.$$

Show that the upper bound for the error in this approximation is less than .

Solution: The error is given by

$$E_1 = \frac{1}{2!} f''(c)(26 - 27)^2 = \frac{1}{2} f''(c)$$

for some $26 \leq c \leq 27$. Now $f''(x) = -\frac{2}{9}x^{-4/3}$ has a negative power of x which is decreasing but then the minus sign outside makes $f''(x)$ increasing. Thus the right endpoint of the interval $[26, 27]$ makes $f''(x)$ largest and we obtain

$$\begin{aligned} f''(x) &= -\frac{2}{9}x^{-4/3} \leq -\frac{2}{9}27^{-4/3} && \text{for } 26 \leq x \leq 27 \\ &= -\frac{2}{9} \cdot \frac{1}{3^4} = -\frac{2}{9 \cdot 81} \end{aligned}$$

So the error is bounded by

$$E_1 \leq -\frac{1}{2} \cdot \frac{2}{9 \cdot 81} = -\frac{1}{729} < -\frac{1}{1000}$$

as required.

4 marks (b)

$$-\frac{1}{1000}$$

4. A tanker off the coast of B.C. breaks in half spilling 10^4 m^3 of oil all at once. The oil floats on the surface and spreads. It has the shape of a cylinder whose height is becoming thinner and thinner. The height, h , of the oil slick is uniform and given by

$$h(t) = \frac{1}{10^4 t}$$

where t is time in hours.

Find the radius of the slick 4 hours later.

Solution: Let t be time measured in hours, r be the radius of the cylinder, and V be the volume of the cylinder. Then

$$V = \pi r(t)^2 h(t).$$

Thus

$$r(t) = \sqrt{\frac{10^4 \cdot 10^4 t}{\pi}} \quad \text{and} \quad r(4) = 10^4 \sqrt{\frac{2}{\pi}} = \frac{20,000}{\sqrt{\pi}}.$$

2 marks (a)

5 marks (b) Find how quickly the slick is spreading across the water surface 4 hours later.

Solution: The quick way is to just differentiate the above expression for $r(t)$ to get

$$r'(t) = \frac{10^4}{2\sqrt{\pi}} t^{-1/2} \quad \text{so} \quad r'(4) = \frac{10^4}{4\sqrt{\pi}} = \frac{2,500}{\sqrt{\pi}}.$$

The long (and tedious) way is to differentiate the volume formula implicitly (treating V as a constant):

$$0 = \pi(2rr'h + r^2h') \quad (*)$$

We need

$$h'(t) = -\frac{1}{10^4 t^2} \quad \text{so that} \quad h'(4) = -\frac{1}{10^4 \cdot 16}$$

Plugging $t = 4$ into equation (*) and dividing by π we get

$$0 = 2 \cdot 10^4 \frac{2}{\sqrt{\pi}} \cdot r' \cdot \frac{1}{10^4 \cdot 4} + \frac{4}{\pi} \cdot 10^8 \cdot \left(-\frac{1}{10^4 \cdot 16}\right) = \frac{r'}{\sqrt{\pi}} - \frac{10^4}{4\pi}$$

and solving for r' :

$$\begin{aligned} \frac{r'}{\sqrt{\pi}} &= \frac{10^4}{4\pi} \\ r' &= \frac{10^4 \sqrt{\pi}}{4\pi} = \frac{10^4}{4\sqrt{\pi}} = \frac{2,500}{\sqrt{\pi}}. \end{aligned}$$

5 marks 5. Naobi is boiling water on the stove to make a cup of coffee. The heating element is at 200°C and the water in her pot is initially at 20°C room temperature. After one minute her water has

reached 40° C. How long will it take for her water to reach the boiling point? Your answer may be left in “calculator-ready” form.

Solution: Let T be the temperature of the can in degree Celsius and t be the time in minutes.

We need to fit the curve $T(t) = Ae^{kt} + E$ to the given data where E is the ambient temperature. In this case the ambient temperature is that of the heating element, i.e., 200° C. $A = T(0) - E = 20 - 200 = -180$. The other data point is $t = 1$ with $T(1) = 40$. Plugging this into the equation we have thus far we get

$$40 = -180e^k + 200$$

Solving for k yields $k = \ln \frac{8}{9}$. Thus

$$T(t) = -180e^{\ln(\frac{8}{9})t} + 200$$

Solving

$$100 = -180e^{\ln(\frac{8}{9})t} + 200$$

for t we find $t = \frac{\ln \frac{5}{9}}{k} \approx 5$ min.